Affine Profile Reduction for Fractional Triangle Packings in Split Graphs

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Abstract

We study fractional triangle packings in split graphs. A split presentation $V(G) = K \sqcup I$ allows the triangles meeting the independent set to be packed explicitly. The unused capacities on the clique edges lead, by finite linear-programming duality, to a triangle-cover problem over one fixed polytope. The objective vector of that cover problem depends affinely on the neighborhood types appearing in I.

The minimum of a linear objective over a fixed polytope is concave in the objective vector. This concavity property reduces the task of proving a uniform lower bound for an arbitrary mixture of independent-set neighborhoods to the analysis of pure neighborhood profiles. After averaging over the stabilizer of the pure profile, the residual cover problem becomes a three-variable linear program. Solving that program gives a uniform quadratic lower bound for the residual value.

As a consequence, every split graph G on n vertices satisfies

$$\boxed{|E(G)| - 2\nu_3^*(G) \le \frac{n^2}{6} + \frac{n}{2}},$$

where $\nu_3^*(G)$ is the maximum fractional triangle-packing value. The leading constant $1/6$ is the one appearing in Erdős Problem #81 on clique partitions of chordal graphs; the present result concerns the finite fractional problem for the subclass of split graphs. The proof is finite and analytic. No asymptotic transfer theorem is used. The complete-split family $K_p \vee \overline{K}_{2p}$ shows that no uniform additive coefficient below $1/6$ is possible. Later papers in the series obtain sharper bounds on the same quantity by independent arguments and do not depend on the present result. What they reuse is the affine reduction and the resulting three-orbit trichotomy.

Keywords: split graph; fractional triangle packing; fractional triangle cover; linear programming; affine reduction; symmetrization.

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1. Introduction

A split graph is a graph whose vertex set can be partitioned into a clique and an independent set. This class is a natural testing ground for clique-partition questions: it retains a rigid complete core while allowing many different neighborhood patterns outside the core. Erdős, Ordman, and Zalcstein [1] exhibited chordal examples requiring asymptotically $n^2/6$ cliques and proved that $(1-c)n^2/4$ cliques suffice for some absolute constant $c > 0$. For split graphs, Chen, Erdős, and Ordman [2] proved the sharper upper bound $\frac{3}{16}n^2 + O(n)$. The present paper studies instead the fractional triangle-packing deficit $|E(G)| - 2\nu_3^*(G)$, for which the result below has leading constant $1/6$. For broader background on clique partitions and coverings, see Cavers [5].

The present paper isolates a fractional mechanism related to the split-graph side of Erdős Problem #81 [3]. Rather than trying to choose edge-disjoint triangles directly, we first build a fractional packing from triangles that meet the independent set. The capacity not used by this first packing remains on the clique edges, and those residual capacities support a second packing formed entirely by clique triangles. Duality then turns this residual packing problem into a fractional triangle-cover problem on the clique.

The main structural point is that the feasible cover polytope does not depend on the independent-set neighborhood profile. Only the objective vector changes. Since the optimum of a minimization problem over a fixed polytope is concave in its objective vector, a mixed neighborhood profile is bounded below by the corresponding convex combination of pure-profile values, and therefore by the smallest of those values. This does not identify a mixed profile with a pure one; it shows that a uniform lower bound may be proved by considering pure profiles. The pure case then has enough symmetry to reduce to three orbit variables.

Role within the broader program. This paper is self-contained apart from the standard finite-dimensional strong linear-programming duality theorem cited and applied in Appendix B. Its purpose is to isolate the affine profile-reduction mechanism in the split-graph setting and to give a finite fractional application. Within the series on Erdős Problem #81, this mechanism is the starting point: Paper II determines the exact fractional maximum of the cover functional over chordal graphs; Paper III proves the linear-error split-graph theorem for clique partitions; Paper IV develops the subsequent chordal bounds and structural stability results [IV]; and Paper V studies colorability and realizability defects through a general transfer framework [V]. None of those later results is used here.

Series notation. In cross-paper discussion, the fractional deficit of Paper I is written $\Phi^*(G) = |E(G)| - 2\nu_3^*(G)$. Paper II writes the same quantity as $\Phi_\tau(G) = |E(G)| - 2\tau_3^*(G)$, while Paper III reserves the unstarred $\Phi(G)$ for the integral deficit $|E(G)| - 2\nu_3(G)$.

The resulting proof has the following shape:

$$\begin{aligned} \text{explicit packing} &\longrightarrow \text{fixed residual polytope} \\ &\longrightarrow \text{affine profile reduction} \\ &\longrightarrow \text{three-orbit optimization} \end{aligned}$$

Our main theorem is finite.

Theorem 1.1 For every split graph G on n vertices,

$$|E(G)| - 2\nu_3^*(G) \le \frac{n^2}{6} + \frac{n}{2}.$$

The accompanying calculation ledger records the exact identities, inequality directions, branch structure, and loss accounting used in the proof. The manuscript keeps the routine algebra visible but compact.

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2. Split-graph notation

Fix a split presentation

$$V(G) = K \sqcup I,$$

where K is a clique and I is independent. Write

$$p := |K|.$$

For $v \in I$, let

$$d_v := |N(v)|.$$

Separate the independent vertices into

$$I_{\geq 2} := \{v \in I : d_v \geq 2\}, \qquad I_1 := \{v \in I : d_v = 1\}, \qquad I_0 := \{v \in I : d_v = 0\}.$$

Put

$$q := |I_{\geq 2}|, \qquad b_1 := |I_1|, \qquad b_{\geq 2} := \sum_{v \in I_{\geq 2}} d_v.$$

Because there are no edges inside I,

$$|E(G)| = \binom{p}{2} + b_{\geq 2} + b_1. \tag{2.1}$$

Also,

$$n = p + q + b_1 + |I_0|. \tag{2.2}$$

Let $\nu_3^*(G)$ denote the maximum value of a fractional triangle packing: a nonnegative weight $w(T)$ is assigned to each triangle T, subject to the condition that the total weight of triangles containing any fixed edge is at most one.

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3. The aggregate load on the clique

For a clique edge $e \in E(K)$, define

$$\kappa_e := \sum_{\substack{v \in I_{\geq 2} \\ e \subseteq N(v)}} \frac{1}{d_v - 1}. \tag{3.1}$$

The normalization is chosen so that each active independent vertex contributes half its degree in total.

Lemma 3.1

$$\sum_{e \in E(K)} \kappa_e = \frac{b_{\geq 2}}{2}.$$

Proof Exchange the order of summation:

$$\begin{aligned} \sum_{e \in E(K)} \kappa_e &= \sum_{v \in I_{\geq 2}} \frac{\binom{d_v}{2}}{d_v - 1} \\ &= \frac{1}{2} \sum_{v \in I_{\geq 2}} d_v \\ &= \frac{b_{\geq 2}}{2}. \end{aligned}$$

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4. A two-phase fractional packing

4.1 Phase I

For each clique edge e, define

$$\lambda_e := \begin{cases} 1, & \kappa_e = 0, \\ \min\{1, \kappa_e^{-1}\}, & \kappa_e > 0. \end{cases}$$

For every triangle vxy with $v \in I_{\geq 2}$ and $x, y \in N(v)$, assign

$$w_1(vxy) := \frac{\lambda_{xy}}{d_v - 1}. \tag{4.1}$$

The load on a clique edge e is

$$\lambda_e \kappa_e = \min\{\kappa_e, 1\}. \tag{4.2}$$

A cross edge vx lies in at most $d_v - 1$ such triangles, each of weight at most $1/(d_v - 1)$. Thus w_1 is feasible.

Define

$$H := \{e : \kappa_e \geq 1\}, \qquad L := \{e : \kappa_e < 1\}.$$

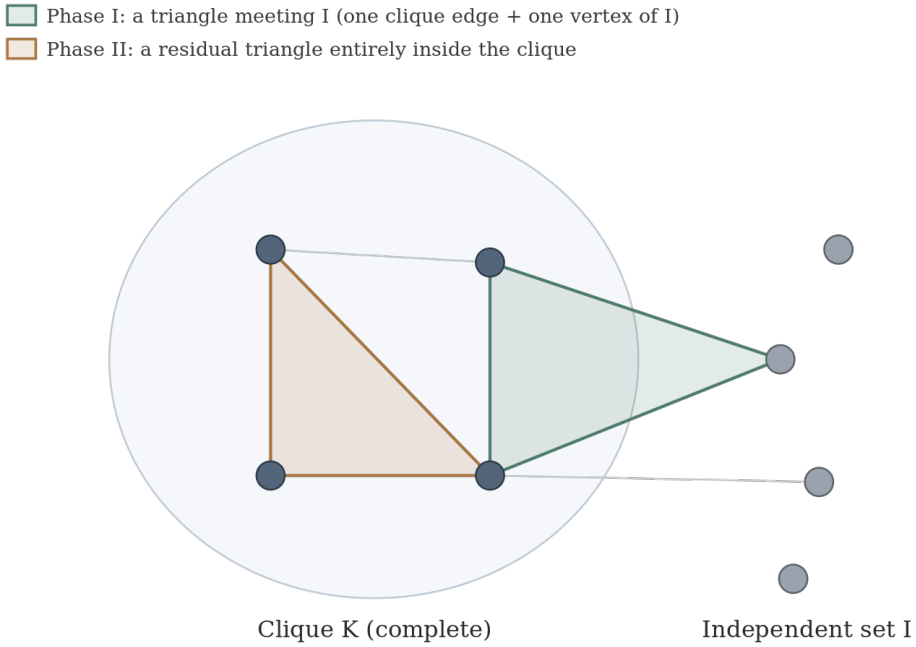


Figure 1: The two-phase fractional packing. Phase I packs triangles meeting the independent set (each uses one clique edge and one vertex of I); the residual capacity remaining on the clique edges then supports Phase II, a packing by triangles lying entirely inside the clique.

Every positive Phase-I triangle contains exactly one clique edge. Hence

$$|w_1| = |H| + \sum_{e \in L} \kappa_e. \tag{4.3}$$

4.2 Phase II

The residual capacity on a clique edge is

$$r_e := \max\{1 - \kappa_e, 0\}. \tag{4.4}$$

Let w_2 be an optimal fractional packing of clique triangles under these residual capacities. The finite packing-covering duality used here is recorded explicitly in Appendix B.

After capping the dual cover variables at one, substitute

$$y_e := 1 - x_e.$$

The triangle-cover constraints become

$$\sum_{e \in E(T)} y_e \leq 2.$$

The coefficients $1 - \kappa_e$ are nonpositive on H, so the maximizing vector may be taken to vanish there. A second substitution,

$$z_e := 1 - y_e,$$

gives $z_e = x_e$ and leads to the fixed capped-dual polytope

$$\mathcal{P}_p := \left\{ z \in [0,1]^{E(K)} : \sum_{e \in E(T)} z_e \geq 1 \text{ for every triangle } T \subseteq K \right\}. \tag{4.5}$$

Define

$$M(\kappa) := \min_{z \in \mathcal{P}_p} \sum_{e \in E(K)} (1 - \kappa_e) z_e. \tag{4.6}$$

Let

$$w_{\text{com}} := w_1 + w_2, \qquad V_{\text{com}} := |w_{\text{com}}|.$$

Here $M(\kappa)$ is the value of a residual cover problem, not the value of an additional packing. Its feasible region depends only on the clique size p, while all information from the independent-set neighborhoods enters through the objective coefficients $1 - \kappa_e$.

The contribution of the edges in H can be accounted for explicitly. Lemma 3.1 and (4.3) give

$$|w_1| = \frac{b_{\geq 2}}{2} + \sum_{e \in H} (1 - \kappa_e).$$

On H, the residual coefficient r_e is zero. In (4.6), by contrast, $1 - \kappa_e \leq 0$; since increasing z_e preserves feasibility, a minimizer may be chosen with $z_e = 1$ on every edge of H. Thus the H-part of $M(\kappa)$ is exactly $\sum_{e \in H}(1 - \kappa_e)$, while its restriction to L is the residual dual value $|w_2|$. Equivalently,

$$|w_2| = M(\kappa) - \sum_{e \in H} (1 - \kappa_e).$$

Adding the two displayed identities gives

$$\boxed{V_{\text{com}} = \frac{b_{\geq 2}}{2} + M(\kappa)}. \tag{4.7}$$

The packing is feasible: on a clique edge, the Phase-I and residual loads add to at most

$$\min\{\kappa_e, 1\} + \max\{1 - \kappa_e, 0\} = 1.$$

Therefore

$$V_{\text{com}} \leq \nu_3^*(G). \tag{4.8}$$

Combining (2.1) and (4.7) yields

$$\boxed{|E(G)| - 2V_{\mathrm{com}} = \binom{p}{2} - 2M(\kappa) + b_1}. \tag{4.9}$$

This cancellation is the bridge from the packing construction to the profile problem.

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5. Affine reduction to a pure neighborhood

For every set $S \subseteq K$ with $|S| \geq 2$, define

$$a_e^S := \frac{\mathbf{1}_{\{e \subseteq S\}}}{|S| - 1}. \tag{5.1}$$

Let n_S be the number of vertices $v \in I_{\geq 2}$ with $N(v) = S$. Then

$$\kappa = \sum_S n_S a^S. \tag{5.2}$$

Assume first that $q > 0$, and set

$$\lambda_S := \frac{n_S}{q}, \qquad \kappa^S := q a^S.$$

Then

$$\kappa = \sum_S \lambda_S \kappa^S, \qquad \lambda_S \geq 0, \qquad \sum_S \lambda_S = 1. \tag{5.3}$$

The reduction follows from an elementary concavity fact.

Lemma 5.1 Let P be nonempty and define the value function

$$m(c) := \min_{x \in P} \langle c, x \rangle.$$

If $c = \sum_i \lambda_i c^{(i)}$ is a convex combination, then

$$m(c) \geq \sum_i \lambda_i m(c^{(i)}) \geq \min_i m(c^{(i)}).$$

Proof For every $x \in P$,

$$\langle c, x \rangle = \sum_i \lambda_i \langle c^{(i)}, x \rangle \geq \sum_i \lambda_i m(c^{(i)}).$$

Take the minimum over $x \in P$.

The residual value from Section 4.2 satisfies $M(\kappa) = m(1 - \kappa)$ when the generic polytope is $P = \mathcal{P}_p$; the different letters distinguish the generic concavity lemma from the manuscript-specific residual notation.

Apply the lemma to the objective vectors $1-\kappa^S$ over the fixed polytope $\mathcal{P}_p$. The conclusion is a lower-bound reduction: a mixed profile need not coincide with any pure profile, but its residual value is at least the corresponding average of the pure-profile values. We obtain

$$\boxed{M(\kappa) \geq \sum_S \lambda_S M(\kappa^S) \geq \min_S M(\kappa^S)}. \tag{5.4}$$

It remains to control a pure profile.

Remark (fixed-polytope concavity) The reduction just used is an instance of a purely convex-analytic fact, stated here in abstract form. Fix a feasible polytope that does not depend on the profile and an objective that depends affinely on the profile mass $\mu = (\mu_S)$. Because the optimal value of a linear program over a fixed polytope is concave in its objective vector, the value at a mixed profile dominates the corresponding convex combination of pure-profile values, and hence their minimum:

$$M_b(\kappa_\mu) \;\geq\; \sum_S \tfrac{\mu_S}{q}\, M_b(\kappa^{S,q}) \;\geq\; \min_S M_b(\kappa^{S,q}).$$

A companion stability estimate bounds the change in the fractional value when the baseline or the profile distribution is perturbed. It is intended for robust and near-split variants. Both are stated for a fixed feasible polytope with an affinely varying objective; neither requires symmetry, which enters only afterwards, on a single pure profile, in Section 6.

6. The pure-profile orbit program

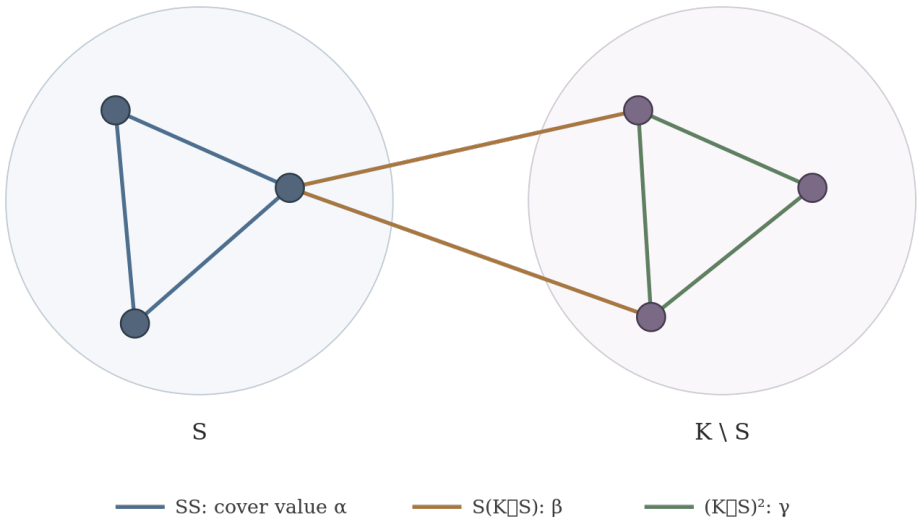


Figure 2: For a pure profile $S \subseteq K$, the clique edges split into three orbits — SS, $S(K \setminus S)$, and $(K \setminus S)(K \setminus S)$ — on which the averaged cover is constant, with values α, β, γ respectively.

Fix $S \subseteq K$, and write

$$s := |S|, \qquad o := p - s.$$

For the pure profile,

$$\kappa_e^S = \begin{cases} q/(s-1), & e \subseteq S, \\ 0, & \text{otherwise.} \end{cases} \tag{6.1}$$

Average an optimal cover over the stabilizer of S. Both the cover polytope and the pure-profile objective are invariant under this group. Averaging therefore preserves feasibility and objective value. The averaged cover is constant on the three edge orbits:

$$SS, \qquad S(K \setminus S), \qquad (K \setminus S)(K \setminus S).$$

Call the corresponding values

$$\alpha, \qquad \beta, \qquad \gamma.$$

The objective is

$$A\alpha + B\beta + C\gamma, \tag{6.2}$$

where

$$A := \frac{s(s-1-q)}{2}, \qquad B := so, \qquad C := \frac{o(o-1)}{2}. \tag{6.3}$$

The existing triangle types impose

$$3\alpha \geq 1, \qquad \alpha + 2\beta \geq 1, \qquad 2\beta + \gamma \geq 1, \qquad 3\gamma \geq 1, \tag{6.4}$$

with any constraint omitted when its triangle type does not exist.

The feasible region is now a three-dimensional polytope. Its relevant extreme points give the uniform, separated, and hot-set covers; when some triangle types do not exist, the corresponding boundary regimes must be treated separately. Solving this three-variable program gives

$$M(\kappa^S) = \begin{cases} \min\{U, D\}, & o \leq 2, \\ \min\{U, D, H\}, & o \geq 3, \end{cases} \tag{6.5}$$

where

$$U := \frac{A + B + C}{3}, \qquad D := A + C, \qquad H := A + \frac{B + C}{3}. \tag{6.6}$$

The three values correspond to the uniform cover, the separated cover, and the hot-set cover.

Table 1. The three-orbit cover program. For a pure profile $S \subseteq K$ with $s = |S|$, $o = p - s$, the residual objective is $A\alpha + B\beta + C\gamma$ over the edge orbits SS, $S(K \setminus S)$, $(K \setminus S)(K \setminus S)$, and its minimum is attained at one of three named covers.

Cover	Value	Orbit values (α, β, γ)
Uniform U	$(A+B+C)/3$	$(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$
Separated D	$A+C$	$(1, 0, 1)$
Hot-set H	$A+(B+C)/3$	$(1, \frac{1}{3}, \frac{1}{3})$

with $A = \frac{s(s-1-q)}{2}$, $B = so$, $C = \frac{o(o-1)}{2}$; and $M(\kappa^S) = \min\{U, D\}$ for $o \le 2$, $\min\{U, D, H\}$ for $o \ge 3$ (Appendix A).

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7. A uniform lower bound for the residual cover

Define

$$R(p,q) := \frac{2p^2 - 2pq - q^2}{12}. \tag{7.1}$$

We compare each candidate in (6.5) with $R(p,q)$.

For the uniform candidate, using $o = p - s$,

$$12(U - R) = q(2o + q) - 2p. \tag{7.2}$$

Since $q, o \ge 0$,

$$U - R \ge -\frac{p}{6}. \tag{7.3}$$

For the separated candidate,

$$12(D - R) = 12o^2 - 6o(2p - q) + (2p - q)^2 - 6p. \tag{7.4}$$

The quadratic part is nonnegative because

$$12u^2 - 6uv + v^2 = 12\left(u - \frac{v}{4}\right)^2 + \frac{v^2}{4}.$$

Thus

$$D - R \ge -\frac{p}{2}. \tag{7.5}$$

For the hot-set candidate,

$$12(H - R) = (2s - q)^2 + 2q(p - s) - 2p - 4s. \tag{7.6}$$

The first two terms are nonnegative, and $s \le p$, so

$$H - R \ge -\frac{p}{2}. \tag{7.7}$$

Consequently,

$$M(\kappa^S) \geq R(p,q) - \frac{p}{2}$$

for every pure profile. By (5.4),

$$\boxed{M(\kappa) \geq R(p,q) - \frac{p}{2}}. \tag{7.8}$$

When $q = 0$, the same estimate follows directly. If $p \leq 2$, then $M(0) = 0$. If $p \geq 3$, the constant cover $z_e = 1/3$ is feasible. To see that it is optimal, sum the cover constraints over all clique triangles: every clique edge occurs in exactly $p - 2$ such triangles, which gives the matching lower bound. Hence

$$M(0) = \frac{p(p-1)}{6}.$$

Remark (where the estimate is tight) The three comparisons above are far from tight except at one configuration. Writing $o = p - s$, the slack $M(\kappa^S) - \big(R(p,q) - p/2\big)$ is at least $9/4$ whenever $s \geq 3$ and $o \geq 3$, and at least $1/4$ in each of the regimes $s = 2$ with $o \geq 1$, $o = 1$, and $o = 2$. Equality occurs only for $o = 0$ together with $q = 2p$: the pure profile whose neighborhood is the whole clique in a graph with $|I| = 2|K|$, which is exactly the complete-split extremizer $K_p \vee \overline{K}_{2p}$ evaluated in Section 9 and identified by Paper II as the terminal family of the fractional extremal problem. Thus the point $(p,q,s) = (2,4,2)$ on the $s = 2$ boundary belongs to the equality family; away from $o = 0$, the $s = 2$ boundary has slack at least $1/4$. Appendix A treats the boundary regimes in full because the minimum of the orbit program genuinely changes form there.

8. Proof of the main theorem

Start from (4.9):

$$|E(G)| - 2V_{\mathrm{com}} = \binom{p}{2} - 2M(\kappa) + b_1.$$

Insert (7.8):

$$|E(G)| - 2V_{\mathrm{com}} \leq \binom{p}{2} - 2R(p,q) + p + b_1. \tag{8.1}$$

A direct simplification gives

$$\binom{p}{2} - 2R(p,q) + p = \frac{(p+q)^2}{6} + \frac{p}{2}. \tag{8.2}$$

Hence

$$|E(G)| - 2V_{\mathrm{com}} \leq \frac{(p+q)^2}{6} + \frac{p}{2} + b_1. \tag{8.3}$$

Write $n = p+q+b_1+|I_0|$ from (2.2), where I_0 is the set of isolated independent vertices. Rather than bounding the quadratic and linear terms of (8.3) separately, subtract (8.3) from the target with sharp quadratic coefficient:

$$\frac{n^2}{6} + \frac{n}{2} - \Big[\frac{(p+q)^2}{6} + \frac{p}{2} + b_1\Big] = \frac{b_1^2 + |I_0|^2 + 2b_1|I_0| + 2(b_1+|I_0|)(p+q) + 3|I_0| + 3q - 3b_1}{6}.$$

Every term on the right is nonnegative except $-3b_1$. If $b_1 = 0$ the inequality is immediate. If $b_1 \geq 1$, then $p \geq 1$, because a degree-one independent vertex has its neighbor in K; grouping $2b_1(p+q) - 3b_1 = b_1(2p+2q-3)$ shows this is nonnegative whenever $p+q \geq 2$. The residual regime is $p = 1,\ q = 0$, where the difference reduces to

$$\frac{b_1(b_1-1) + |I_0|^2 + 2b_1|I_0| + 5|I_0|}{6} \geq 0,$$

with equality exactly when $b_1 = 1$ and $|I_0| = 0$. Hence

$$|E(G)| - 2V_{\text{com}} \leq \frac{n^2}{6} + \frac{n}{2}. \tag{8.4}$$

Finally, $V_{\text{com}} \leq \nu_3^*(G)$. Thus

$$|E(G)| - 2\nu_3^*(G) \leq |E(G)| - 2V_{\text{com}} \leq \frac{n^2}{6} + \frac{n}{2}.$$

This proves Theorem 1.1.

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9. Discussion

Three elementary ideas drive the proof. We first separate the packing into a part supported on triangles meeting the independent set and a residual part supported entirely on the clique. We then formulate the residual problem over one fixed cover polytope. Symmetry is used only after the lower-bound problem has been reduced to pure profiles.

The fixed-polytope feature is the central structural point. Concavity applies because the feasible region does not change when the neighborhood profile changes; only the objective vector does. If the constraints themselves depended on the profile, the reduction in Section 5 would no longer follow from the same argument. Orbit averaging is a separate step: it uses the symmetry of a pure profile and is not needed for the affine reduction itself.

The intermediate quantities p, q, and κ depend on the chosen split presentation. The final estimate does not. Once the terms are assembled, the conclusion is expressed only in terms of $n = |V(G)|$.

The additive term $n/2$ in Theorem 1.1 is not optimized. For the complete-split family $K_p \vee \overline{K}_{2p}$, with $p \geq 2$ and $n = 3p$, every triangle uses at least one clique edge, so $\nu_3^*(G) \leq \binom{p}{2}$, while the Phase-I construction attains equality. Consequently, the left-hand side of Theorem 1.1 equals $n^2/6 + n/6$ on this family, so the optimal additive term lies between $n/6$ and the $n/2$ proved here. Determining it exactly is outside the scope of the present argument.

The paper is intentionally limited to the finite fractional inequality. Any passage to integral triangle packings, clique partitions, asymptotic consequences, or wider graph classes requires additional results and belongs to a separate stage.

The reusable component is the fixed-polytope affine reduction, which is independent of the particular finite bound proved here.

Paper III of the series later reuses the same pure-profile three-orbit computation in an integral setting. With its notation $d = s$ and $p - d = o$, the three branches of its common-profile value $F(p, q, d)$ are $qd/2 + U$, $qd/2 + D$, and $qd/2 + H$, where $qd/2$ is the Phase-I contribution of the pure profile in the present paper; its term $C_\alpha p^2$, for $\alpha = q/p$, is the comparison value $R(p, q)$ of Section 7. This is a forward cross-reference only: no result of Paper III is used here.

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10. Formalization and reproducibility

The accompanying formalization and reproducibility document identifies the unchanged Lean/Mathlib freeze, its theorem interfaces and the recorded build and axiom evidence. This editorial edition changes neither the mathematical results nor the frozen sources; no Lean/Lake rebuild was performed. Computational checks are supplementary evidence, not premises of the proofs.

Appendix A. The three-orbit program and its boundary cases

This appendix supplies the details behind (6.5). It is included because the cases in which one of the four triangle types does not exist are easy to obscure in a compressed orbit calculation.

For a pure profile $S \subseteq K$, write

$$s := |S|, \qquad o := p - s,$$

and let α, β, γ denote the cover values on the edge orbits

$$SS, \qquad S(K \setminus S), \qquad (K \setminus S)(K \setminus S).$$

The objective is

$$A\alpha + B\beta + C\gamma,$$

where

$$A = \frac{s(s - 1 - q)}{2}, \qquad B = so, \qquad C = \frac{o(o - 1)}{2}.$$

Only constraints corresponding to existing triangle types are imposed.

A.1 The interior case $s \geq 3$ and $o \geq 3$

All four constraints are present:

$$3\alpha \geq 1, \qquad \alpha + 2\beta \geq 1, \qquad 2\beta + \gamma \geq 1, \qquad 3\gamma \geq 1.$$

If $A < 0$, decreasing the objective favors the largest feasible value of α, so an optimum has $\alpha = 1$. The remaining lower boundary joins

$$(\beta, \gamma) = (0, 1)$$

and

$$(\beta, \gamma) = \left(\frac{1}{3}, \frac{1}{3}\right).$$

Because the objective is linear on that segment, its minimum is attained at an endpoint. These two endpoint values are

$$D = A + C$$

and

$$H = A + \frac{B + C}{3}.$$

If $A \geq 0$, then for fixed β the least feasible values of α and γ are

$$\alpha = \gamma = \max\left\{\frac{1}{3}, 1 - 2\beta\right\}.$$

For $0 \leq \beta \leq 1/3$, the objective is linear between the separated point and the uniform point, with values D and

$$U = \frac{A + B + C}{3}.$$

For $\beta \geq 1/3$, the objective is

$$U + B\left(\beta - \frac{1}{3}\right) \geq U,$$

because $B \geq 0$. Hence the minimum is one of U, D, H.

A.2 The case $s = 2$

The SSS triangle type does not exist, so the constraint $3\alpha \geq 1$ is omitted.

If $A < 0$, the same monotonicity argument gives $\alpha = 1$, and the endpoint analysis above remains valid.

If $A \geq 0$ and $o \geq 3$, the missing constraint introduces one additional boundary endpoint at $\beta = 1/2$. Its objective value exceeds the uniform value by

$$\frac{B - 2A}{6}.$$

For $s = 2$,

$$B - 2A = 2o - 2(1 - q) = 2(o + q - 1) \geq 0$$

whenever this endpoint is relevant. Therefore it does not improve on U. The cases $o = 0, 1, 2$, in which some triangle types and their constraints are absent, are treated separately in A.3; the excess formula above is not invoked there.

A.3 The cases $o = 0, 1, 2$

When $o = 0$, we have $B = C = 0$. If $s \geq 3$, the only triangle constraint is $3\alpha \geq 1$, so

$$\min_{\alpha \in [1/3,1]} A\alpha = \min\left\{\frac{A}{3}, A\right\} = \min\{U, D\}.$$

If $s = 2$, then $p = 2$ and there are no clique-triangle constraints. Since the pure-profile case has $q > 0$, we have $A = 1 - q \leq 0$, and the minimum over $0 \leq \alpha \leq 1$ is $A = D$. Thus the formula $\min\{U, D\}$ remains valid.

When $o = 1$, there are no $(K \setminus S)^2$ edges and no triangle type entirely in $K \setminus S$. The reduced feasible region is an interval, and its endpoints give U or D.

When $o = 2$, the triangle type entirely in $K \setminus S$ does not exist. A possible extra endpoint has

$$(\beta, \gamma) = \left(\frac{1}{2}, 0\right)$$

and objective value

$$A + \frac{B}{2}.$$

Since $o = 2$,

$$\frac{B}{2} = s \qquad \text{and} \qquad C = 1,$$

so

$$A + \frac{B}{2} \geq A + C = D.$$

Thus this endpoint is dominated by D.

Consequently,

$$M(\kappa^S) = \begin{cases} \min\{U, D\}, & o \le 2, \\ \min\{U, D, H\}, & o \ge 3. \end{cases}$$

No nonexistent orbit or triangle constraint is used in any boundary regime.

Appendix B. Finite packing-covering duality

For completeness, we record the specialized finite packing-covering dual pair used in Section 4, prove weak duality, verify the hypotheses required for the finite-dimensional strong linear-programming duality theorem cited below, and apply that theorem to the residual program.

Proposition B.1 Let $\mathcal{T}$ be a finite family of triangles, let E be the finite set of edges appearing in them, and let $r = (r_e)_{e \in E}$ be a nonnegative capacity vector. Then

$$\max\left\{ \sum_{T \in \mathcal{T}} w_T : w_T \ge 0,\ \sum_{T \ni e} w_T \le r_e \text{ for every } e \in E \right\}$$

equals

$$\min\left\{ \sum_{e \in E} r_e x_e : x_e \ge 0,\ \sum_{e \in E(T)} x_e \ge 1 \text{ for every } T \in \mathcal{T} \right\}.$$

Both optima are attained.

Proof Let A be the edge-triangle incidence matrix, so that $A_{eT} = 1$ when $e \in E(T)$ and $A_{eT} = 0$ otherwise. The two programs are

$$\max\{\mathbf{1}^{\mathsf{T}} w : Aw \le r,\ w \ge 0\}$$

and

$$\min\{r^{\mathsf{T}} x : A^{\mathsf{T}} x \ge \mathbf{1},\ x \ge 0\}.$$

Weak duality follows directly. Indeed, for every feasible pair w, x,

$$\mathbf{1}^{\mathsf{T}} w \le x^{\mathsf{T}} A w \le x^{\mathsf{T}} r.$$

The primal feasible region is nonempty, since $w = 0$ is feasible. It is also bounded in every coordinate: if $T \in \mathcal{T}$, then for each edge $e \in E(T)$,

$$0 \le w_T \le \sum_{T' \ni e} w_{T'} \le r_e.$$

Hence the primal optimum is finite and attained. The dual is feasible, for example with $x_e = 1$ for every e. Finite-dimensional strong linear-programming duality [4, Chapter 7] therefore gives equality of the two attained optimum values. Applying the proposition to the clique triangles and the residual capacities r_e in (4.4) yields the dual program used in Section 4.

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